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Individual Assignment

1. Set Up Windows Workgroup System

1.1 What is Windows Workgroup System?

A Windows Workgroup is a logical grouping of networked computers that share resources (files, printers) on a peer-to-peer basis. In this model, there is no central server managing the network. Each computer maintains its own local database for security, meaning user accounts and passwords are managed individually on each machine. It is typically used in small environments (homes or small offices) where centralized administration is not required.

1.2 How is Windows Workgroup System differing from Windows Active Directory Domain System?

Feature | Windows Workgroup | Active Directory (Domain)

Administration | Decentralized. Each computer is managed separately. | Centralized. Managed via Domain Controllers.

Security | Low. Users manage their own security settings. | High. Consistent security policies applied globally (Group Policy).

User Accounts | Users must have an account on each computer they access. | A single user account can log in to any computer in the domain.

Scalability | Best for small networks (typically under 10–20 devices). | Scalable to thousands of users and devices.

DNS Dependency | Does not require internal DNS. Uses NetBIOS/LLMNR. | Heavily relies on DNS for locating domain resources.

1.3 Steps in Configuring Windows Workgroup System for 6 Users in Windows 10

To set up a Workgroup and add 6 distinct users:

Step A: Configure the Workgroup Name

Right-click This PC > Properties > Advanced system settings.

Click the Computer Name tab > Change.

Under "Member of," select Workgroup and type a name (e.g., IT_DEPT). Click OK.

Restart the computer.

Step B: Create 6 User Accounts

Go to Settings > Accounts > Family & other users.

Click Add someone else to this PC.

Select I don't have this person's sign-in information > Add a user without a Microsoft account.

Enter the Username (e.g., User1) and Password.

Repeat this process 5 more times for User2 through User6.

1.4 Steps for File Sharing in Windows Workgroup

Navigate to the folder you wish to share using File Explorer.

Right-click the folder and select Properties.

Click the Sharing tab and select Advanced Sharing.

Check Share this folder.

Click Permissions.

Add specific users and assign permissions.

Ensure Network Discovery is turned on.

1.5 Steps for Printer Sharing in Windows Workgroup

Go to Settings > Devices > Printers & scanners.

Select the printer and click Manage.

Click Printer properties > Sharing.

Check Share this printer.

Assign a Share name.

2. Set Up Windows Active Directory Domain System

2.1 What is Windows Active Directory Domain System?

Windows Active Directory Domain Services (AD DS) is a directory service for Windows domain networks. It stores information about network objects and manages authentication and authorization.

2.2 & 2.3 Installation Steps and Definitions for Windows Server 2022 AD DS

Step 1: Open Server Manager

Manage > Add Roles and Features.

Step 2: Installation Type

Choose Role-based or feature-based installation.

Step 3: Server Selection

Select the local server from the server pool.

Step 4: Server Roles

Check Active Directory Domain Services and add required features.

Step 5: Confirmation and Install

Install and then Promote this server to a domain controller.

Step 6: Deployment Configuration

Select Add a new forest. Enter domain name (e.g., dbu.edu.et).

Step 7: Domain Controller Options

Keep DNS and Global Catalog selected.

Set a DSRM password.

Step 8: Paths and Review

Choose storage paths (default recommended).

Install and reboot.

2.4 Steps to Create AD Users and User Groups

Creating a User Group:

Open Active Directory Users and Computers.

Create an Organizational Unit (OU) named IT_Students.

Right-click and create a Group named Network_Admins.

Creating an AD User:

Right-click IT_Students OU > New > User.

Fill in user details.

Add user to Network_Admins group.

Appendix: List of Figures

Figure 1: Server Manager Dashboard – Starting the "Add Roles and Features" Wizard.

Figure 2: Select Server from the Server Pool.

Figure 3: Selecting "Active Directory Domain Services" Server Role.

Figure 4: Deployment Configuration – Adding a New Forest.

Figure 5: Domain Controller Options (DNS and Global Catalog).

Figure 6: Creating a New Organizational Unit (OU).

Figure 7: Creating a New Security Group inside the OU.